A

TDI CAPSTONE PROJECT

DOCUMENTATION

ON

DATA ANALYSIS, INTERPRETATION AND VISUALIZATION USING SQL AND MICROSOFT EXCEL.

BY

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1.0 INTRODUCTION

Spire Pro Integrated Limited is a Tech. company that manufactures quality hardware and software, and provides services for various fields including cloud computing, database management, ERP, customer relationship management and business intelligence. Services. Recently, I have been employed as a data analyst and assigned to investigate the high employee turnover rate at our company. Using comprehensive data from the HR department, I am tasked with analysing factors like age, job, race, and tenure and its effect on attrition in the company.

1.2. PROBLEM STATEMENT

The company is experiencing high employee turnover, negatively impacting productivity, morale, and operational costs. Understanding the primary factors contributing to attrition is critical for developing targeted strategies to reduce turnover rates.

1.3. PROJECT OBJECTIVES

- ❖ To analyse the impact of demographics on attrition.
- ❖ To identify key factors contributing to employee attrition.
- ❖ To gain insights into trends and patterns in employee turnover.
- ❖ To evaluate the effect of tenure on employee turnover.
- ❖ To provide data-driven recommendations.

1.4. RESEARCH QUESTIONS

a) What impact do employee demographics such as age, gender, and race have on attrition rates within the company?
   i) How do different demographic groups compare in terms of turnover?
   ii) Are certain demographics more prone to leaving the company?

b) What are the key factors contributing to employee attrition?
   i) Which aspects of the work environment, job roles, or personal factors are most strongly associated with employees deciding to leave?

c) How does the length of tenure affect the likelihood of an employee leaving the company?
   i) Is there a critical tenure period after which employees are more likely to leave?

   ii) How does turnover vary between new hires and long-term employees?

d) What trends and patterns can be identified in employee turnover over time?
   i) Are there specific periods during the year when turnover is higher?
   ii) How has the attrition rate changed over the past few years?

e) What strategies can be implemented to reduce employee turnover?

2.0 DATA SOURCE & IMPORTATION

The data source was from The Data Initiative (TDI). I renamed the dataset and saved it as a csv.txt file before importing it to SSMS. To understand the data, I examined the data structure within the SQL Server and also profiled the data to gain a deep understanding of the structure of the data and how to go about the data cleaning process using the following queries respectively;

```
SELECT TOP 10 * FROM
[Capstone Dataset];

SELECT
    COUNT(*) AS TotalRecords,
    COUNT(DISTINCT id) AS UniqueIDs,
    AVG(DATEDIFF(YEAR, birth_date, GETDATE())) AS AverageAge,
    MIN(hire_date) AS FirstHireDate,
    MAX(hire_date) AS LastHireDate,
    SUM(CASE WHEN term_date IS NULL THEN 1 ELSE 0 END) AS currently employed
FROM [Capstone Dataset];
```

2.1. DATA TRANSFORMATION

2.1.1. DATA CLEANING AND PROCESSING

I started the data transformation processing with cleaning and pre-processing, this was done to ensure the dataset for analysis is accurate and consistent. These are the steps I took to achieve a clean dataset.

- **Handling missing values:** I used the coalesce function to search for missing values.
- **Removing duplicates:** I identified them first, and then reviewed them before removing duplicates.

❖ **Data Standardization:** This involved formatting to ensure data is consistent, it included; fixing gender values, identifying incorrect spellings in job titles and updating them, date formatting, setting names to proper case, and removing leading or trailing spaces.

For the pre-processing, I started by creating new variables like tenure years, age, employment status, department headcount, average tenure by department, gender ratio, average age by job title, attrition status and retention rate using the ALTER TABLE statement. (Please see appendix for all queries).

I proceeded to restructure the data to ensure it was suitable for analysis and integration by running queries to get my columns for tenure years, age, attrition status, average tenure by department and job title, monthly and yearly hires, and monthly and yearly attrition.

2.2. EXPLORATORY DATA ANALYSIS AND QUERYING

a) **Attrition rate by department:** This was queried to determine the percentage of employees who have left based on their department. I used the select and count statements for the department and terminated date respectively to arrive at this, see the query below;

```
SELECT
department,
COUNT(*) AS total_employees,
COUNT(CASE WHEN term_date IS NOT NULL THEN 1 END) AS num_terminated,
(COUNT(CASE WHEN term_date IS NOT NULL THEN 1 END) * 100.0 / COUNT(*))
AS attrition_rate
FROM [HR2 Dataset]
GROUP BY department;
```

b) **Attrition rate by job title:** this query was run to identify if they are high-risk roles laden with stress or demand for specific sets of skills. It will also help in identifying job satisfaction levels. The job title was selected and the count statement was applied to the terminated date to get the attrition rate per job title. The query used was thus;

```
SELECT job_title,
(COUNT(CASE WHEN term_date IS NOT NULL THEN 1 END) * 100.0 / COUNT(*))
AS attrition_rate
FROM [HR2 Dataset]
GROUP BY job_title
ORDER BY attrition_rate DESC;
```

c) **<u>Attrition Rate by Tenure:</u>** This analysis was imperative to gain insight into how long an employee stays before leaving the company. The same process used above was applied but by tenure;

```
SELECT
    tenure_years,
    (COUNT(CASE WHEN term_date IS NOT NULL THEN 1 END) * 100.0 /
COUNT(*)) AS attrition_rate
FROM [HR2 Dataset]
GROUP BY tenure_years ORDER BY tenure_years;
```

d) **<u>Attrition Rate by Gender</u>:** This was necessary to help with gender-specific trends contributing to employee turnover. This analysis was necessary to understand whether certain genders are more likely to leave the company.

```
SELECT
gender,
(COUNT(CASE WHEN term_date IS NOT NULL THEN 1 END) * 100.0 / COUNT(*))
AS attrition_rate
FROM [HR2 Dataset]
GROUP BY gender;
```

e) **<u>Attrition Rate by Race/Ethnicity</u>:** This will help identify if there are any disparities between the racial and ethnic groups, bringing to light discrimination or work culture. The same select and count statement was used to query this;

```
SELECT race,
(COUNT(CASE WHEN term_date IS NOT NULL THEN 1 END) * 100.0 / COUNT(*)) AS
attrition_rate
FROM [HR2 Dataset]
GROUP BY race
ORDER BY attrition_rate DESC
```

f) **<u>Retention Rate:</u>** This is the percentage of employees who remain with the company over a specific period. This analysis is crucial for understanding how well the company retains employees over time.

```
SELECT
(COUNT(CASE WHEN term_date IS NULL THEN 1 END) * 100.0 / COUNT(*)) AS
retention_rate
FROM [HR2 Dataset];
```

After all the analysis and queries were run, I backed up my data and exported my saved file to Excel to begin data visualization.

3.0 DATA VISUALIZATIONS

For visualisation, I created sheets for my visuals and segmented them as follows; attrition by demography, attrition by department, attrition by job title, attrition by tenure, dashboard and automated reports.

3.1. Attrition by Demography

A) AGE: I created pivot tables from the dataset to show the relationship between age, gender, location and attrition rate as well as the employee distribution across each demographic metric. A clustered column chart was used to represent the information as shown below;

Tab. 1. Distribution of Employees by Age Group

Age Group	Count of Employee ID	Sum of Attrition Flag
20-29	4860	563
30-39	6142	732
40-49	5894	677
50-59	5318	600
Grand Total	**22214**	**2572**

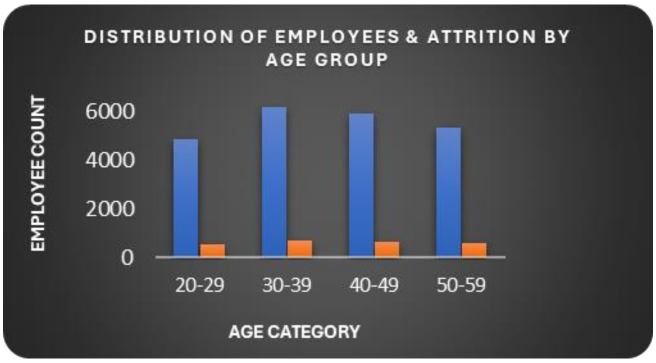

Fig 1. Distribution of Employees and Attrition by Age Group

B) GENDER: I used a clustered bar chart to show the distribution of employees according to their genders and the relationship between them and attrition.

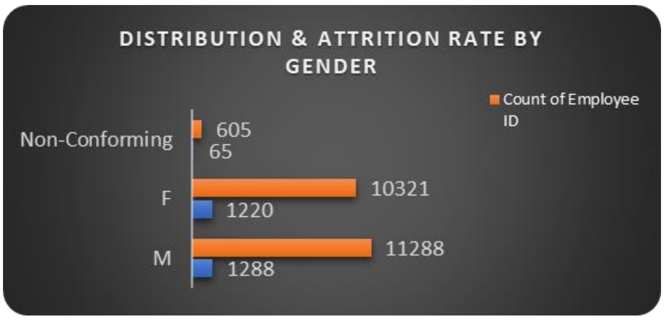

Fig 2. Distribution & Attrition Rate by Gender

C) LOCATION: The options from the data were given as employees working remotely and stationed at headquarters. I used a pivot chart to show this relationship with attrition and the distribution across both locations, For the visuals, I used a heatmap on the pivot chart to show this as seen below;

Tab.2. Distribution of Employees & Attrition by Location

Location	Count of Employee ID	Sum of Attrition Flag
Headquarters	16715	1968
Remote	5499	604
Grand Total	**22214**	**2572**

D) CITY: I used a pivot chart to create this, with about 77 cities making up the workforce, I applied filters to show the top 10 cities with the highest number of employees.

Tab.3. Top 10 Distribution of Employees by City

City	Count of Employee ID
Chicago	351
Cincinnati	285
Cleveland	16871
Dayton	202
Detroit	203
Indianapolis	200
Lexington	199
Louisville	228
Philadelphia	326
Pittsburgh	293
Grand Total	**19158**

E) RACE/ETHNICITY: To show the data visuals of this, I created a pivot chart and using conditional formatting, I gradient filled with bars to pinpoint the race with the highest attrition, I also created a pie chart to show this visual. (See in Summary of findings).

Tab.4. Distribution of employees & attrition by Ethnicity

Race/Ethnicity	Count of Employee ID	Sum of Attrition Flag
American Indian or Alaska Native	1327	149
Asian	3562	418
Black or African American	3619	424
Hispanic or Latino	2501	269
Native Hawaiian or Other Pacific Islander	1229	159
Two or More Races	3648	425
White	6328	728
Grand Total	**22214**	**2572**

3.2. Attrition by Department:

To show the visuals, a pivot chart was generated to show the number of attritions per department. A clustered column chart was used to show the visuals.

Tab. 5. Attrition by Department

Department	Sum of Attrition Flag
Engineering	787
Accounting	384
Human Resources	212
Sales	210
Training	208
Services	188
Business Development	167
R & D	131
Support	113
Product Management	73
Marketing	48
Legal	42
Auditing	9
Grand Total	**2572**

To gain further insights, I created another pivot chart to establish the relationship between each department, city and attritions. I used conditional formatting to create a gradient bar chart to pinpoint which city and department exactly is experiencing the highest number of attritions.

Tab.6. Heatmap of Attrition by City and Department

Count of Attrition Flag	City					
City	**Chicago**	**Cincinnati**	**Cleveland**	**Philadelphia**	**Pittsburgh**	**Grand Total**
Accounting	61	39	2498	58	49	2705
Auditing	1		43		1	45
Business Development	28	17	1243	21	27	1336
Engineering	84	98	5118	102	66	5468
Human Resources	43	33	1352	25	30	1483
Legal	1	3	253	2	4	263
Marketing	7	2	385	9	2	405
Product Management	8	11	487	7	7	520
R & D	19	12	807	20	18	876
Sales	32	25	1375	25	27	1484
Services	27	15	1292	28	27	1389
Support	13	8	730	9	15	775
Training	27	22	1288	20	20	1377
Grand Total	**351**	**285**	**16871**	**326**	**293**	**18126**

3.3. Attrition by Job Title

After creating the pivot chart, it showed there were about 185 job titles, therefore the filter option was introduced to show the top 6 job titles with the highest attrition rate. This visual was represented using a pie chart.

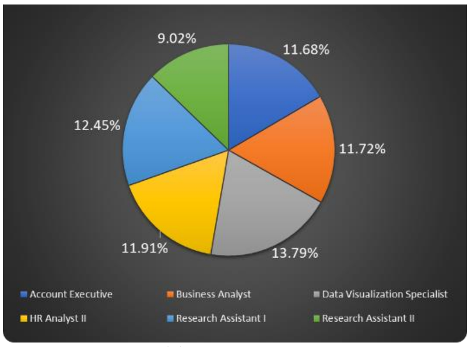

Fig 3. Top Six (6) Attrition Rate by Job Title

I also created a pivot table showing the top 8 job roles with the most terminations. I used a clustered bar chart to show this visual.

Fig 4. Top 8 Terminated Count by Job Title

3.4. Attrition by Tenure

I created a visual for this using the line chart with markers to show the trend over time. I also used the pivot table to get the average tenure of employees in the company.

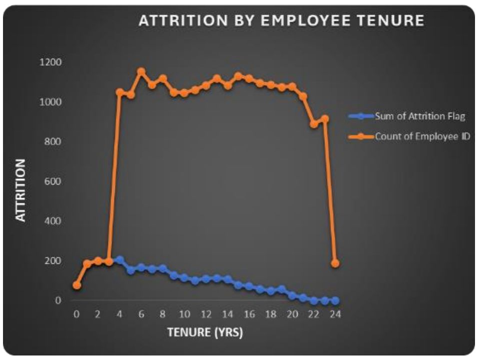

Fig 5. Attrition by Employee Tenure

3.5. Retention Rate

I created a pivot table and used a clustered bar chart to show the department with highest retention rate and their age categories. This will give insight to the age groups with the highest retention rate.

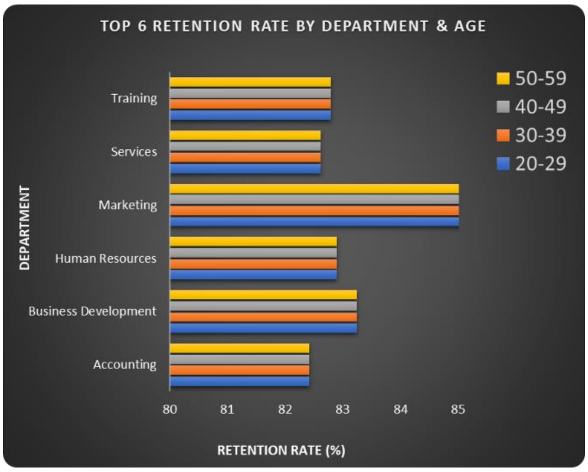

Fig 6. Top 6 Retention Rates by Department & Age

4.0 SUMMARY OF FINDINGS

4.1. IMPACT OF DEMOGRAPHICS ON ATTRITION

4.1.1 AGE

- **High Attrition Rate in Younger Age Groups**

  The 20-29 and 30-39 age brackets have slightly higher attrition rates compared to older age groups. This could be due to younger employees seeking better opportunities, career advancement, or further education.

- **Stable Attrition in Middle Age Groups**

  The 40-49 and 50-59 age groups show relatively stable and slightly lower attrition rates. Employees in these age groups might have more settled careers and are less likely to leave for significant reasons.

- **Overall Consistent Attrition Rate**

  The overall attrition rate across all age groups is relatively consistent, indicating that attrition is not heavily skewed towards any particular age group.

4.1.2 GENDER

- **Higher Attrition Rate for Females**

  The attrition rate for females (11.82%) is slightly higher than that for males (11.40%) and non-conforming individuals (10.74%). This indicates that female employees are leaving the company at a higher rate compared to their male and non-conforming counterparts.

- **Non-Conforming Employees**

  Although the number of non-conforming employees is relatively small, their attrition rate is also significant. This suggests that there may be specific challenges faced by non-conforming employees that need to be addressed.

4.1.3 LOCATION

- **Higher Attrition at Headquarters**

  The attrition rate at the headquarters (11.77%) is slightly higher than the overall attrition rate (11.58%) and the remote attrition rate (10.98%). This suggests that employees working at the headquarters are leaving at a higher rate compared to remote employees.

- **Lower Attrition for Remote Employees**

  The attrition rate for remote employees is lower than that for headquarters employees. This could indicate that remote work arrangements are more favourable for employee retention, possibly due to better work-life balance, flexibility, and reduced commuting stress.

4.1.4 CITY

- **High Attrition in Cincinnati and Lexington**

  Cincinnati (12.98%) and Lexington (12.56%) have the highest attrition rates among the cities listed. This indicates potential issues in these locations that need to be addressed.

- **Lower Attrition in Detroit and Philadelphia**

  Detroit (8.87%) and Philadelphia (9.51%) have the lowest attrition rates, suggesting that employees in these cities are more satisfied or face fewer issues leading to turnover.

- **Significant Attrition in Cleveland**

Cleveland has the highest number of employees and a significant attrition rate (11.74%), indicating that even small percentage changes can greatly impact overall attrition numbers.

4.1.5 RACE/ETHNICITY

- **High Attrition Among White Employees**

  The highest number of attrition instances is among White employees (6328), indicating a significant impact on overall attrition rates.

- **Significant Attrition in Minority Groups:**

  Black or African American (3648), Hispanic or Latino (3619), and Asian (3562) employees also show high attrition numbers, suggesting potential issues related to workplace inclusivity or support.

- **Lower Attrition in Smaller Groups**

  Native Hawaiian or Other Pacific Islander (1229) and American Indian or Alaska Native (1327) have lower attrition numbers, but these groups may still face unique challenges that need to be addressed.

4.2 ATTRITION BY DEPARTMENT

- **High Attrition in Engineering**

  Engineering has the highest number of attrition instances (787), indicating significant turnover in this department. This could be due to various factors such as job stress, lack of career growth, or a competitive job market.

- **Moderate Attrition in Accounting and Human Resources**

  Accounting (384) and Human Resources (212) also show notable attrition numbers. These departments may face challenges related to workload, job satisfaction, or compensation.

- **Lower Attrition in Specialized Departments**

  Departments like Auditing (9), Legal (42), and Marketing (48) have lower attrition numbers, suggesting better retention in these areas. This could be due to specialized skills, job satisfaction, or effective management practices.

4.3 IMPACT OF JOB ROLES ON ATTRITION

- **High Attrition in Key Roles**

  Roles like Business Analyst, HR Analyst II, and Data Visualization Specialist have high attrition rates, indicating potential issues such as job dissatisfaction, high stress, or lack of career growth.

- **Moderate Attrition in Technical Roles**

  Technical roles such as Software Engineer I and Desktop Support Technician show moderate attrition, suggesting a need for better support and development opportunities.

- **Significant Attrition in Research and HR Roles**

  Research Assistant I and HR Analyst II also show high attrition rates, suggesting challenges in these roles that need to be addressed.

- **Low Attrition in Specialized Roles**

  Specialised roles like Account Coordinator and Assistant Professor have low attrition, indicating higher job satisfaction or fewer opportunities for turnover.

4.4 IMPACT OF LENGTH OF TENURE ON ATTRITION

- **High Initial Attrition**

  The highest attrition occurs within the first two years of employment. This suggests that new hires may face challenges in adapting to the company culture, job expectations, or work environment.

- **Decreasing Attrition Over Time**

  Attrition rates decrease significantly after the initial years, indicating that employees who stay beyond the first few years are more likely to remain with the company long-term.

- **Stable Long-Term Retention**

  Employees with longer tenure (11+ years) show minimal attrition, suggesting strong loyalty and satisfaction among long-term employees.

4.4 RETENTION RATE

- **Consistent Retention Rates Across Age Groups**

  The retention rates for each department are consistent across all age categories (20-29, 30-39, 40-49, 50-59). For example, Accounting has a retention rate of 82.42% across all age groups.

- **Highest Retention Rates**

  The Marketing department shows the highest retention rate across all age groups, consistently at 85.43%.

- **Lowest Retention Rates:** Human Resources has the lowest retention rate at 82.9%.

- **Overall Retention Rate**

  The overall retention rate for the company is approximately 82.86%, with slight variations by age group, showing 82.87% for ages 20-39 and 82.85% for ages 40-59.

5.0 CONCLUSION

Based on the analysis, several key factors impacting employee attrition have been identified:

a) Location-Based Attrition

Headquarters has a slightly higher attrition rate **(11.77%)** compared to remote employees **(10.98%).** This suggests that employees at the headquarters may face more challenges, possibly related to workplace culture or commuting stress.

b) City-Based Attrition:

Cincinnati (12.98%) and **Lexington** (12.56%) have the highest attrition rates among the cities listed, indicating potential issues specific to these locations.

Detroit (8.87%) and **Philadelphia** (9.51%) have the lowest attrition rates, suggesting better retention practices or more favourable working conditions in these cities.

c) Department-Based Attrition:

Engineering has the highest number of attrition instances (787), indicating significant turnover in this department. This could be due to job stress, lack of career growth, or a competitive job market.

- Departments like **Accounting** and **Human Resources** also show notable attrition numbers, suggesting challenges related to workload or job satisfaction.

4. Attrition by Job Title:

- High attrition rates are observed in roles such as **Business Analyst** (11.72%), **HR Analyst II** (11.91%), and **Data Visualization Specialist** (13.79%). These roles may face issues related to job dissatisfaction, high stress, or lack of career growth.

5. Attrition by Employee Tenure:

- The highest attrition occurs within the first two years of employment, indicating challenges in the initial adaptation period for new hires.
- Attrition rates decrease significantly after the initial years, suggesting that employees who stay beyond the first few years are more likely to remain with the company long-term.

6. Retention by Age Category and Department:

- Retention rates are consistent across different age categories within each department, indicating stable retention practices.
- **Marketing** has the highest retention rate (85.43%), suggesting effective retention strategies in this department.

6.0 RECOMMENDATIONS

- ❖ **Develop tailored retention strategies**: for high attrition roles and locations as well as different age groups based on their specific needs and motivations. This could include career development opportunities, mentorship programs, competitive compensation packages and addressing specific local issues.
- ❖ For example, younger employees might value career progression, while older employees might prioritize job security and benefits.
- ❖ **Enhance Onboarding and Training Programs:** Implement comprehensive onboarding and training programs to help new employees integrate smoothly into the company. Provide clear job expectations and support during the initial years to reduce early attrition.

- **Improve Workplace Culture and Environment**: Focus on creating a positive and inclusive workplace culture, especially at the headquarters. Consider offering flexible work arrangements, such as hybrid models, to reduce stress and improve job satisfaction.
- **Leverage Best Practices from Low-Attrition Areas**: Identify and replicate successful retention strategies from departments and locations with low attrition rates. For example, the practices used in the Marketing department, which has high retention, could be applied to other areas.
- **Continuous Monitoring and Feedback:** Regularly monitor attrition rates and gather feedback from employees to understand their needs and concerns. Use this data to make informed decisions and adjust retention strategies to ensure continuous improvement.